

ETA0506 - Computer vision
class Test Assessment Tool-1

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1. Image enhancement in spatial domain.
Definition

Image enhancement in the spatial domain improves image quality by directly modifying pixel

a) Gray-level Transformations.

Modify pixel intensity values.

Techniques: negative, log, power-law.

[gamma], contrast stretching.

use. Improves brightness and contrast.

b) Histogram processing.

Improves image contrast by redistributing gray levels.

c) spatial filtering.

uses a mask (kernel) to process neighboring pixels.

Smoothing filters: mean, Gaussian, median.

Sharpening filters: Laplacian, Sobel, unsharp Masking

justification

Gray-level transformation $\rightarrow$ Brightness adjustment.

histogram processing $\rightarrow$ Better Contrast.

spatial filtering $\rightarrow$ noise removal and sharper edges.

Application:

*) Medical imaging, satellite images, CCTV surveillance, Document enhancement.

Conclusion:

*) spatial domain enhancement improves image quality by enhancing brightness, important details.

2. Image Smoothing Image vs Image Sharpening.

Image smoothing image sharpening
removes noise enhances edges blurs image makes
clearer uses low-pass filters uses high-pass.

Removes frequency components enhances
high-frequency components.

Image smoothing:

smoothing reduces noise and unwanted
details. It uses low-pass filters.

working:

A filter mask is moved over the image.
pixel values are calculated using neighbouring
pixels.

eg: mean filter, Median filter.

Application:

noise removal, medical image preprocessing

Image sharpening:

sharpening enhances edges and fine
details. It uses high-pass filters.

working:

Details sudden changes in intensity.
enhances edges and boundaries.
makes the image appear.

Eg: Laplacian, Sobel, Prewitt.

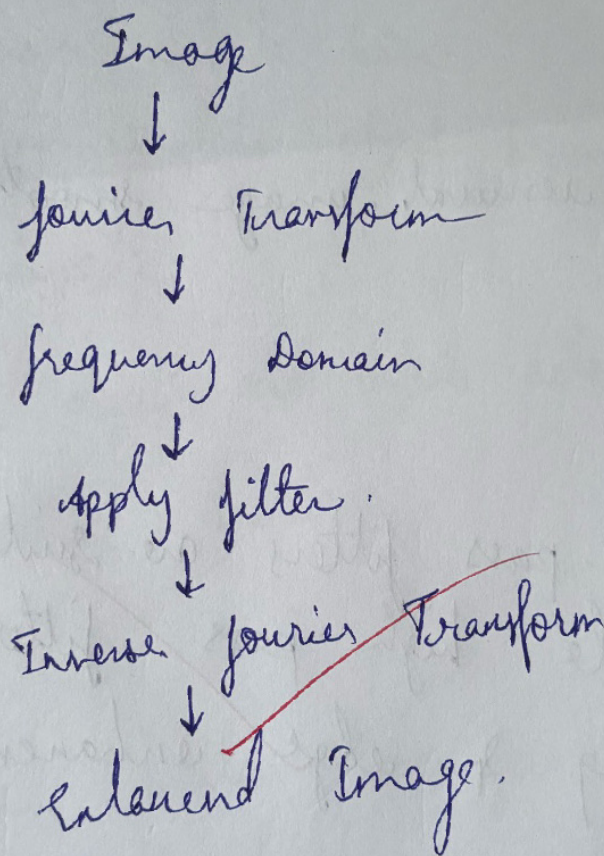
Conclusion:

Smoothing is used when noise needs to be removed.

3. Image Enhancement In frequency Domain

Introduction:

process:



low-pass filter:

A low-pass filter allows low frequencies and reduces high frequencies.

effects:

- Removes noise.
- Reduces fine details
- Smooths the image
- Can cause blurring.

Applications:

noise removal, image smoothing, preprocessing.

Conclusion:

low-pass filters are suitable for smoothing, while high-pass filters are suitable for sharpening & edge enhancement.

4. Image segmentation:

Definition:

Image segmentation is the process of dividing an image into meaningful region or objects.

Types:

point Detection: Detects isolated points.

line Detection: Detects lines.

edge Detection: Detects object boundaries.

Suitable Technique:

for object and defect detection, edge detection with the three holding.

Reason:

edge detection identifies the objects boundary, while thresholding separates.

Conclusion:

segmentation helps in object recognition. defect detection.

5. Edge Detection, Edge Linking and Boundary Detection.

Introduction:

Edges are important features that represent sudden changes in intensity. Edge detection, linking, and boundary detection are important for recognising objects and defects.

1. Edge Detection:

Edge Detection identifies sudden intensity changes in an image.

Eg: Sobel, Prewitt, Roberts, Canny.

Dark Region | Bright Region
↑
edge.

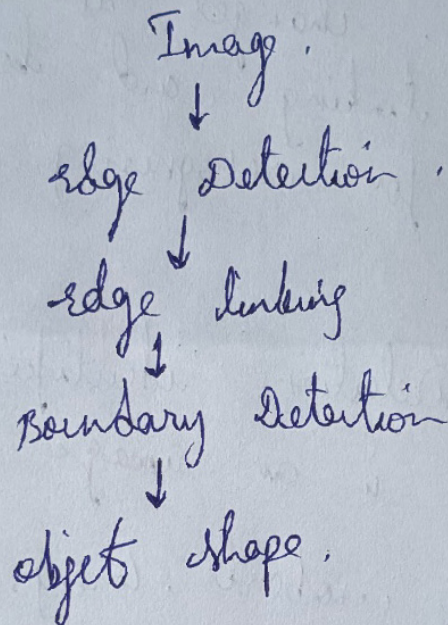
2. Edge Linking:

⇒ Edge linking connects broken or disconnected edge pixels.

broken edges → edge linking → continuous

3.7 Boundary Detection:

Boundary detection identifies the complete outline of an object.



Conclusion:

*> edge detection finds edges, edge linking connects them, and boundary detection forms the complete object boundary.